

Technical Learning Report: Day 4

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Subject: Test Planning

1. Objective

The objective of the day was to develop a practical understanding of **test planning** and its major components, including test plan structure, test strategy, test scope, entry and exit criteria, risks and assumptions, estimation, resources, schedule, responsibilities, and deliverables. The practical task was to prepare a comprehensive and risk-aware test plan for a **Member Application** using an IEEE-style test plan structure. The plan was designed to identify what would be tested, how testing would be performed, what conditions must be satisfied before and after testing, and which risks should receive priority based on their business impact.

2. Technical Breakdown

A. Test Planning

Test planning involves defining the overall objectives, scope, approach, resources, activities, risks, and schedule required for testing an application.

The prepared Member Application test plan documented:

- Test objectives and purpose.
- Test items and features to be tested.
- Features that were not included in testing.
- Test approach and test levels.
- Test types and test techniques.
- Test data requirements.
- Entry and exit criteria.
- Suspension and resumption conditions.
- Test deliverables.
- Testing tasks.
- Environmental requirements.
- Responsibilities.
- Staffing and training requirements.
- Testing schedule.
- Risks and contingencies.

- Approvals.

The test plan was organized using a structured IEEE-style test documentation format.

B. Test Items and Scope

The Member Application test plan identified the major features included within the testing scope:

- Member registration and login.
- Password reset.
- Member profile management.
- Membership subscription and payment.
- Membership renewal and cancellation.
- Member directory and search.
- Notification emails.
- Admin panel.
- Member management and reports.

The plan also clearly identified features outside the testing scope, including the internal processing logic of the third-party payment gateway and the hosting provider's own uptime guarantees.

This helped establish a clear boundary between what the testing team would verify and what would remain outside the application's testing responsibility.

C. Test Strategy

The test approach defined multiple levels and types of testing.

Test levels:

1. Component testing
2. Integration testing
3. System testing
4. User Acceptance Testing

The integration level included areas such as API and payment gateway integration, while system testing covered the complete application behavior. User Acceptance Testing involved real member representatives.

The planned test types included:

- Functional testing.
- Usability testing.
- Security testing.
- Performance testing.
- Compatibility testing.

The plan also identified testing techniques such as equivalence partitioning, boundary value analysis, decision tables, and exploratory testing. Test data included anonymized or synthetic member records and sandbox payment cards.

D. Entry and Exit Criteria

Entry criteria were defined as conditions that should be satisfied before testing begins.

The identified entry criteria were:

- Requirements and acceptance criteria are approved.
- Test environment and test data are ready.
- Build is deployed and passes the smoke test.

Exit criteria defined the conditions for completing the planned testing:

- All planned high- and medium-risk test cases are executed.
- No open Critical or High severity defects remain.
- Defined coverage of requirements and risk items is achieved.

E. Scope, Risks and Assumptions

The test plan considered the application's important business areas, particularly membership registration, authentication, payments, renewals, member information, notifications, and administration.

The plan also distinguished between product risks and project risks. Risks were evaluated using:

Risk Level = Likelihood × Impact

Business impact was used to justify the impact rating instead of relying only on technical assumptions. The risk analysis considered consequences such as:

- Revenue loss.
- Member complaints.
- Loss of trust.
- Legal or compliance exposure.

- Poor member experience.
- Release delays.
- Increased project cost.

The document also included project risks such as delayed payment gateway sandbox access and limited team experience with payment integrations.

F. Estimation and Schedule

The test plan included a structured testing schedule:

Activity	Start	End
Test planning & case design	Week 1	Week 2
Component & integration testing	Week 3	Week 4
System testing	Week 5	Week 6
UAT	Week 7	Week 7
Test completion report	Week 8	Week 8

This provided a planned sequence for the testing activities and showed how different testing stages would be distributed across the project timeline.

G. Resources, Responsibilities and Environment

The test plan identified the required testing resources and responsibilities.

Staffing:

- 2 testers.
- 1 test lead.

Training was identified for the payment gateway sandbox tool before integration testing.

Responsibilities were distributed among:

- Test Manager.
- Testers.
- Developers.
- Product Owner / Business Representative.

The required testing environment included:

- Staging environment.
- Payment gateway sandbox account.
- Chrome, Safari, and Firefox on desktop.
- Android and iOS mobile devices.

H. Suspension and Resumption Criteria

The test plan also defined when testing should be temporarily stopped.

Testing would be suspended when:

- The testing environment is unavailable for more than four hours.
- A blocking defect prevents further test execution.

Testing would resume after the blocking issue was fixed and confirmed in a new build.

3. Practical Implementation

A. Member Application Test Plan

A complete test plan was prepared for the Member Application using a structured IEEE-style format.

The document included:

- Test Plan Identifier.
- References.
- Introduction and Purpose.
- Test Items.
- Features Not to Be Tested.
- Test Approach.
- Item Pass/Fail Criteria.

- Suspension and Resumption Requirements.
- Test Deliverables.
- Testing Tasks.
- Environmental Needs.
- Responsibilities.
- Staffing and Training Needs.
- Schedule.
- Risks and Contingencies.
- Approvals.

B. Test Strategy Implementation

The practical test plan defined the testing strategy by specifying:

- Test levels.
- Test types.
- Test techniques.
- Test data.
- Testing sequence.

The strategy covered the flow from component testing through integration and system testing to User Acceptance Testing.

C. Entry and Exit Criteria Implementation

Entry and exit conditions were defined for the Member Application.

Before testing, requirements, acceptance criteria, environment, test data, and the deployed build were required to be ready.

For completion, high- and medium-risk test cases had to be executed, Critical and High severity defects had to be closed, and the defined requirements/risk coverage had to be achieved.

D. Risk Register

A detailed risk register was created containing **nine identified risks**, covering both product and project risks.

The risks included:

1. Payment processing failure or double charging.
2. Exposure of member personal/payment data.
3. Missing or delayed renewal reminder emails.

4. Login failure during peak load.
5. Incorrect or incomplete member directory search.
6. Unauthorized membership-status changes through the admin panel.
7. Broken mobile layout.
8. Delayed payment gateway sandbox access.
9. Limited experience with payment integration testing.

Each risk included likelihood, impact, risk level, business impact justification, and a corresponding testing or mitigation action.

E. Business-Based Risk Justification

The risks were not evaluated only from a technical perspective. Each risk was connected to its possible business consequence.

For example, payment-processing failure was identified as a **High** risk because it could result in direct revenue loss, complaints, chargebacks, and loss of trust.

Similarly, security weaknesses were considered high impact because they could result in legal penalties, reputational damage, and loss of member trust.

This demonstrated the use of **business priority when deciding testing risk levels and priorities**.

F. Test Deliverables and Tasks

The test plan identified the expected testing deliverables:

- Test cases and test scripts.
- Test progress reports.
- Defect reports.
- Test completion report.

Testing tasks included preparing test cases from requirements and risk analysis, setting up the test environment and test data, executing test levels, tracking defects to closure, and reporting status to stakeholders.

4. Key Findings & Best Practices

- A test plan provides a structured approach for defining how testing will be performed.
- Testing scope should clearly identify both features to be tested and features not to be tested.

- Test strategy should define appropriate test levels, test types, techniques, and test data.
- Entry criteria help ensure that testing starts only when the required conditions are ready.
- Exit criteria provide measurable conditions for determining when planned testing can be completed.
- Risks should be evaluated using likelihood and impact.
- Business impact provides an important basis for prioritizing testing risks.
- High-risk business functions such as payment and security require stronger testing attention.
- Project risks such as environment or third-party dependency delays can also affect testing progress.
- A test plan should define responsibilities, resources, environment, deliverables, and schedule in addition to test activities.
- Suspension and resumption criteria provide defined conditions for handling situations that prevent normal testing.
- Risk mitigation actions can be included directly within the test plan to connect identified risks with testing activities.

5. Conclusion

The day's work provided practical understanding of **test planning and risk-aware testing** through the preparation of a complete Member Application test plan. The work covered test scope, test strategy, test levels and types, entry and exit criteria, test deliverables, testing tasks, environment, responsibilities, staffing, schedule, and suspension/resumption conditions. A detailed risk register was also prepared with nine risks, where each risk was evaluated according to likelihood and impact and supported by a business impact justification. This helped demonstrate how business priority can influence testing focus and risk management. Overall, the practical work resulted in a comprehensive IEEE-style test plan for the Member Application with a clearly defined testing approach and business-focused risk management.